

20250513EB_MN (NCT,ATP,TDP43, IF)

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2025年5月13日 星期二

Differentiation: BP w/ 1.5mM pyruvate (later stage 0.5mM pyruvate)
Experiment: T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A + glutamax, b-mercaptoethanol, N2, B27)
split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年5月15日 星期四

T2 -> T3

2025年5月17日 星期六

T3 -> T4 (~14 mL for over-weekend)

2025年5月20日 星期二

T4 -> T5

2025年5月22日 星期四

T5 -> T6

2025年5月24日 星期六

T6 -> T6 (~14mL for over-weekend)

2025年5月27日 星期二

- T6 -> T7
Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 2. Remove the PLO and wash the plate with PBS once
 3. Coat the plate with laminine for 1hr.
 4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P16+9	8

Table1 ^

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6

Table3 ^

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	1.36	99	5	6.8	A	24	283	24	500

Well1 ^

	1	2	3	4	5	6
A	111		92		no virus	
B						
C						
D	110					

2025年5月28日 星期三

Change T7 -> T7 if added virus

2025年5月30日 星期五

T7 -> BP T8 (with N2, B27, P/S, additional 1mM sodium pyruvate)

2025年5月31日 星期六

-300 +500 BP T8 (with N2, B27, P/S, additional 1mM sodium pyruvate)

2025年6月2日 星期一

-300 +500 BP T8 (with N2, B27, P/S)
*Newly prepared BP with N2, B27, P/S w/o additional sodium pyruvate

According to the original protocol (<https://pmc.ncbi.nlm.nih.gov/articles/PMC4443325/>)
There is 0.5mM glutamine in BP already, do not need to add
They didn't add b-mercaptoethanol as supplement
The recommended range for pyruvate is <1mM.
This [paper](#) mentioned that *optimal neuroprotection was achieved with relatively low concentrations of pyruvate (≤ 1 mm), whereas at high concentration (10 mm) pyruvate was ineffective. This paradox could result from the cytosolic acidification induced by the cotransport of pyruvate and protons into neurons.*

2025年6月4日 星期三

-300 +500 BP T8 (with N2, B27, P/S)

Imaging vLH110, 111

Well2							^
	1	2	3	4	5	6	
A	1	2	A	H	1	2	
B	4	3	B	G	4	3	
C	1	2	C	F	1	3	
D	4	3	D	E			

media preparation_SirDNA							
	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	SirDNA dilution	SirDNA volume
2	1	2.5	2000			5000	0.4
3	2	0.15	2000			5000	0.4
4	3	0	2000			5000	0.4
5	4	0	2000	2	6	5000	0.4

Table5						
	A	B	C	D	E	F
1			110		111	
2	number	condition	treatment time	image 1h	treatment time	image 1h
3	1	2.5mM	12:50	13:50	10:20	11:30
4	2	0.15mM	13:40	14:40	10:50	12:03
5	3	0mM	14:20	15:22	11:20	12:26
6	4	0mM+FCCP+2DG	15:00	16:00	12:00	13:00

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	*111_2,3,4 &110_3 uses 5%
4	488	16	50	1s	

10 positions/well (10 positions would be the max. there is only 2 sec between loops)

Used 16 bit instead of 12 bit (the S/N is much better!!!!)

110: pre-activation: 3 loops, 30s interval; activation: 30 loops, 30s interval

111: pre-acitvation: 3 loops, 30s interval; activation: **25 loops**, 30s interval

vLH92

prepare media with different conc. of glucose1

	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	A	1100	20	55	2000	36.4	1963.6
3	B	20	7.5	2.6666666667	1000	375.0	625.0
4	C	20	2.5	8	1600	200.0	1400.0
5	D	2.5	0.5	5	1000	200.0	800.0
6	E	2.5	0.15	16.6666666667	1000	60.0	940.0
7	F	2.5	0.05	50	1000	20.0	980.0
8	G	2.5	0.015	166.6666666667	1000	6.0	994.0
9	H		0				

O.C. , 40X, zoom1.5, 600x900, 16-bit

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	300
3	Dapi-ex-Green-em	16	20	500

- Timelapse Imaging
- 1. mark positions (16:00-17:40)
 - 2. Pre-treatment: 3 loops (17:40-18:00)
 - 3. Post-treatment: 7 loops (18:05-19:15)

2025年6月6日 星期五

TDP43 (A56, B56)

Prepare 1st dilution: Halo ligand (stock: 1mM) to 2.5uM: $1000/2.5=400$.

Prepare 399uL DMSO with 1uL 1mM Halo dye.

Use leftover pre-diluted 2.5uM JFX650.

positions: 10 soma, 8 neurite movie (500ms, 40 loops)

Table4

	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	Halo ligand (2.5uM -> 1nM)	SPY595 (2500x)	FCCP	2DG
2	1	2.5	1000	0.4	0.4		
3	2	0.15	1000	0.4	0.4		
4	3	0	1000	0.4	0.4		
5	4	0mM+FCCP+2DG	1000	0.4	0.4	1	3.0

TDP43 exp time stamp

	A	B	C	D
1	number	condition	treatment time	image 1h
2	1	2.5mM	10:30	11:33
3	2	0.15mM	11:00	12:12
4	3	0mM	11:45	12:48
5	4	0mM+FCCP+2DG	12:20	13:20

O.C. , 60X, 600x900, 16-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	500ms	
3	640	16	20	500ms	
4	640 movie	16	40	500ms	40 loops

IF (C56, D56)

1. Treat C56 with condition 1 and 3 respectively at 2:05pm
2. remove media from the plate
3. wash with cold PBS at 3:05pm
4. fix cells with 4% PFA for 10 min, RT
5. wash with PBS
6. permeabilize cells with 0.2% Triton-X for 10 min
7. Blcoking with 10% FBS --> cold rooom over weekend

2025年6月9日 星期一

primary antibody o/n

Table2								
	A	B	C	D	E	F	G	H
1		antibody	species	dilution			total volume	Ab volume
2	C-32	Anti-MNX1 (HB9) Antibody	Rb	100	spinal motor neuron fate markers	D5	200	2
3	c-28	Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	m	200	pan-neuronal markers	D5	200	1
4	121	VACHT	M	500	motor neuron marker	D5	200	0.4
5	C.13	MAP 2	Rb	500	pan-neuronal markers	D6	200	0.4
6	33	TDP-43	Rb	250		C5,6	400	1.6
7	121	VACHT	M	500	motor neuron marker	C5&6	400	0.8